

PRANAV ROY

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Education

Vellore Institute of Technology

Bachelor of Technology in Computer Science and Engineering

Bhopal, Madhya Pradesh

2023 - 2027

Kendriya Vidhyalaya

Higher Secondary CBSE

Sector 47, Chandigarh

2020 - 2022

Experience

WebInOrbit

Software Engineer Intern

May 2026 - Present

- **Full-Stack & Backend Engineering:** Architected **Clinic Cloud**, a production-grade Hospital Management System (**20+ modules**) using **Java 22, Spring Boot**, with scalable RESTful APIs built on a layered Controller-Service-Repository architecture featuring DTO mapping, Bean Validation, **JWT Auth**, role-based access control (RBAC), and frontend-backend API integration.
- **Database & Performance Optimization:** Designed a normalized **PostgreSQL** schema with complex entity relationships, **JPA/Hibernate** mappings, cascading operations, indexing, and optimized fetching (Lazy/Eager Loading), improving transactional consistency, scalability, and API performance by 20%.
- **Business Workflow Automation:** Implemented ACID-compliant **@Transactional** workflows for patient management, billing, prescriptions, inventory sync, and payment/refunds, ensuring data integrity across modules.
- **Real-Time Communication & Frontend Development:** Integrated secure **WebRTC**-based doctor-patient teleconsultation and built a responsive frontend with **Next.js**, **React**, **TypeScript**, **React Hook Form**, and **Zod** for state management and validation.

Projects

Event Microservice Platform [🔗 Link](#) | NestJS, Apache Kafka, Redis, Docker, PostgreSQL, Drizzle ORM **July 2026**

- Developed a distributed **event-driven microservices architecture** by decoupling business logic into Auth, Events, Tickets, and Notification services.
- Implemented asynchronous inter-service communication using **Apache Kafka** for scalable, fault-tolerant event processing.
- Built a production-grade backend using **NestJS, PostgreSQL**, and **Drizzle ORM** with type-safe database access.
- Designed a centralized **API Gateway** featuring global exception filters, response interceptors, and request validation for consistent API behavior.
- Created a shared internal library for reusable DTOs, interfaces, and utilities, while integrating **Redis** for rate limiting and containerizing services using **Docker Compose**.

Streamly [🔗 Link](#) | WebRTC, Next.js, Node.js, Express.js, JavaScript, Tailwind CSS, REST APIs **December 2025**

- Built a full-stack real-time streaming platform using **Next.js**, **TypeScript**, and **Tailwind CSS** on frontend, with **Node.js** and **Express.js** for backend.
- Implemented low-latency **WebRTC** with signaling and media stream management for real-time audio/video communication.
- Integrated the **MediaRecorder API** to enable client-side recording and efficient storage of live streaming sessions.
- Developed backend REST APIs to manage session lifecycle, recording metadata, and media-related operations, ensuring efficient request handling.
- Used **Next.js App Router**, server-side rendering, and optimized routing to deliver a responsive cross-device UI/UX.

Awards & Achievements

- Ranked Globally 230 in LeetCode Biweekly Contest 156 out of 17,717 participants.
- Solved 100+ DSA questions across various platforms.

Technical Skills

Languages: C++, JavaScript, TypeScript, SQL.

Technologies: React.js, Node.js, REST APIs, Git, PostgreSQL, Redis, Docker, AWS EC2/S3 Linux.

Testing & Tools: Postman, GitHub Actions.

Core Competencies: Object-Oriented Programming, Problem Solving, Agile, Database Design, Data Structures & Algorithms.